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# D. Product of Binary Decimals

time limit per test: 3 seconds
memory limit per test: 256 megabytes
input: standard input
output: standard output

Let's call a number a *binary decimal* if it is a positive integer and all digits in its decimal notation are either $0$ or $1$. For example, $1\,010\,111$ is a binary decimal, while $10\,201$ and $787\,788$ are not.

Given a number $n$, you are asked whether or not it is possible to represent $n$ as a product of some (not necessarily distinct) binary decimals.

## Input

The first line contains a single integer $t$ ($1 \le t \le 5 \cdot 10^4$) — the number of test cases.

The only line of each test case contains a single integer $n$ ($1 \le n \le 10^5$).

## Output

For each test case, output "YES" (without quotes) if $n$ can be represented as a product of binary decimals, and "NO" (without quotes) otherwise.

You can output "YES" and "NO" in any case (for example, strings "yES", "yes", and "Yes" will be recognized as a positive response).

## Example

**input** Copy

```
11
121
```

Codeforces Round 937 (Div. 4)
Finished
Practice

→ Virtual participation

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Start virtual contest

→ Submit?

Language: Java 21 64bit
Choose file: Choose File No file chosen
Submit

```
1
14641
12221
10110
100000
99
112
2024
12421
1001
```

**output** Copy

```
YES
YES
YES
YES
YES
YES
NO
NO
NO
NO
YES
```

## Note

The first five test cases can be represented as a product of binary decimals as follows:

- $121 = 11 \times 11$.
- $1 = 1$ is already a binary decimal.
- $14\,641 = 11 \times 11 \times 11 \times 11$.
- $12\,221 = 11 \times 11 \times 101$.
- $10\,110 = 10\,110$ is already a binary decimal.

## → Problem tags

dp implementation number theory

No tag edit access

## → Contest materials

- Announcement (en)
- Tutorial (en)

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